PRACTICAL- 15

Final Integration, Testing, Documentation, GitHub Submission, and Local Deployment

AIM / Problem Definition: Integrate all modules into a single deployable StudentHub portal. Verify navigation, responsiveness, authentication, database operations, security checks, AJAX functionality, and documentation. Deploy locally using XAMPP/Apache virtual host and submit GitHub repository.

Tools / Technology Used: Full-Stack Web Technologies: HTML5, CSS3, JavaScript (ES6+), PHP 8.x, MySQL, Apache Virtual Hosts, Git / GitHub, XAMPP

Project Implementation & Architecture

Project Overview (StudentHub — Final Integration, Testing, Documentation, GitHub Submission, and Local Deployment)

Practical 15 represents the capstone culmination of the StudentHub University Portal. It integrates all 14 preceding practical modules into a unified, secure, deployable web application. This practical includes comprehensive end-to-end testing via an automated diagnostic runner (test-suite.php), formal verification reporting (test-report.html), security auditing, and local Apache Virtual Host deployment.

Technical Architecture & Workflow

End-to-End System Integration: Unifies all static HTML5 pages, dynamic PHP scripts, RESTful JSON APIs, database models, session guards, and responsive stylesheets into a coherent navigation flow.

Automated System Diagnostic Test Runner (test-suite.php): A live in-browser automated verification tool that runs 10 systematic diagnostics across environment, database connectivity, seed data volume, BCRYPT hashing, CSRF security, session configurations, and API endpoints.

Comprehensive Formal Test Report (test-report.html): Detailed documentation covering Course Outcomes (CO1-CO5) achievement matrix, practicals 1-15 test execution logs, security compliance checklist, and accessibility audits.

Web Application Security Hardening: Verification of defense-in-depth: SQL injection prevention via prepared statements, XSS defense via htmlspecialchars, CSRF tokens with hash_equals(), BCRYPT password hashing, session regeneration, and .htaccess upload shields.

Apache Virtual Host Deployment (Intermediate Extension): Step-by-step configuration of local DNS mapping and Apache httpd-vhosts.conf enabling seamless deployment under custom local domain http://studenthub.local.

Output Verification & Functional Screens

Output 1: Integrated StudentHub Landing Page & Navigation

Fully responsive university portal landing page with carousel banner, interactive theme switcher, FAQ accordions, and unified navigation linking all practical modules.

Output 2: Automated Diagnostic Test Runner (test-suite.php)

In-browser diagnostic runner reporting 10/10 PASS score with execution timings, green status badges, and system environment specifications.

Output 3: Course Outcome (CO1–CO5) Achievement Matrix

Tabular verification matrix in test-report.html mapping curriculum competencies to completed practical implementations with 100% achievement status.

Output 4: End-to-End Test Execution Logs (Practicals 1–15)

Comprehensive testing table documenting inputs, expected results, actual outputs, and pass confirmations across all 15 practical assignments.

Output 5: Security Compliance Audit Checklist

Verification of parameterized queries, XSS sanitization, CSRF token validation, BCRYPT password hashing, and upload directory protection.

Output 6: Local Apache Virtual Host Deployment (http://studenthub.local)

Successful browser access under custom virtual host domain http://studenthub.local verifying production-like local hosting.

Key Questions & Technical Analysis

Q1. Are all pages properly linked and navigable?

Yes. Navigation headers, breadcrumbs, and footers across all static and dynamic pages have been standardized with consistent routing, active page indicators, and zero broken links.

Q2. Are sessions, roles, validations, and database operations working end-to-end?

Yes. The entire user lifecycle—from public exploration, user registration with BCRYPT hashing, role-based login redirection, student/admin CRUD operations, AJAX APIs, to secure session teardown—functions flawlessly end-to-end.

Q3. Are security and accessibility checklists completed?

Yes. The application conforms to WCAG 2.1 AA accessibility guidelines (semantic HTML5, ARIA labels, skip-to-content links, color contrast) and satisfies comprehensive security standards against SQL injection, XSS, CSRF, and session hijacking.

Q4. Is project documentation sufficient for installation and testing?

Yes. Detailed documentation is provided in README.md, test-report.html, and the database schema dump database.sql, providing step-by-step setup instructions for XAMPP and Apache Virtual Hosts.

Q5. Is the project repository clean and properly committed?

Yes. The Git repository is maintained with meaningful, descriptive commit history, semantic file naming (no practical1/practical2 aliases), and pushed to https://github.com/smit-2406/studenthub-website.

GitHub Repository Link

GitHub Repository URL: https://github.com/smit-2406/studenthub-website

Conclusion & Learning Outcomes

Practical 15 successfully verified the complete integration, robust security, automated testing, and local deployment of the StudentHub University Portal. The full-stack application fulfills all Course Outcomes (CO1 through CO5) and demonstrates professional engineering practices from requirement analysis to production-ready deployment.